

Leveraging AlphaFold2 Predictions for Structure-Based Drug Discovery: Opportunities and Limitations

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Abstract

AlphaFold2 has dramatically expanded the structural coverage of the human proteome, enabling structure-based drug discovery for previously undruggable targets. We analyze 1,247 AlphaFold2 structures of disease-relevant proteins and assess their utility for virtual screening campaigns. While high-confidence regions (pLDDT > 90) show docking performance comparable to experimental structures, low-confidence disordered regions and holo-form prediction remain significant challenges. We identify 89 cryptic binding sites predicted by AlphaFold2 that are absent in available crystal structures, suggesting new druggable opportunities. The conformational flexibility required for induced-fit docking in these sites remains an open problem.

1. Introduction

Drug discovery typically requires high-resolution 3D protein structures to enable rational design of small molecule inhibitors. Historically, structural data has been available for only ~30% of human proteins with therapeutic relevance. The 2021 release of AlphaFold2 (Jumper et al., 2021) and its subsequent database covering 200M+ proteins represents a paradigm shift, providing predicted structures for virtually every protein in the human proteome.

However, critical questions remain about the reliability of these predictions for drug discovery. Predicted structures represent apo-form (ligand-free) conformations, whereas binding sites often undergo conformational changes upon ligand binding (induced fit). Additionally, intrinsically disordered regions, which are common in transcription factors and signaling proteins, receive low confidence scores and may not represent meaningful binding geometries.

2. Methods

We downloaded AlphaFold2 structures for 1,247 proteins linked to OMIM disease entries with no existing approved drugs. Structures were processed using Schrödinger's Protein Preparation Wizard. Binding site prediction used FPocket 4.0 and SiteMap.

For validation, we compared AlphaFold2-based docking (Glide SP) against experimental crystal structure docking for 203 proteins with both available. Ligand libraries included ZINC15 (2M drug-like compounds) and FDA-approved drugs (2,650 compounds). Docking accuracy was assessed by re-docking known co-crystallized ligands and measuring RMSD.

3. Binding Site Analysis

Among 1,247 analyzed proteins, 847 (67.9%) had at least one predicted binding pocket with SiteScore > 0.8. Of these, 89 proteins showed cryptic sites - pockets present in the AlphaFold2 structure but absent in all available PDB entries for that protein. Cryptic sites were enriched in protein-protein interaction interfaces (41%) and allosteric regions (33%).

High-confidence regions (pLDDT > 90) covering predicted binding sites correlated strongly with experimental binding site geometry (RMSD < 1.5 Å in 78% of cases). Low-confidence regions (pLDDT < 70) showed poor geometry preservation, with 61% having RMSD > 3 Å.

4. Virtual Screening Performance

Re-docking of co-crystallized ligands into AlphaFold2 structures achieved < 2 Å RMSD in 71% of cases for high-confidence binding sites, compared to 84% for experimental structures. For kinases, performance was near-equivalent (79% vs 82%). For GPCRs, performance dropped sharply (52% vs 77%), likely due to transmembrane helix flexibility.

Prospective virtual screening of the FDA drug library against 50 novel targets identified 131 candidate interactions. Of 24 experimentally tested, 9 (37.5%) showed binding affinity $K_d < 10^{-6}$ M, validating AlphaFold2-based screening as a useful but imperfect filter.

Notably, allosteric sites in transcription factors showed the lowest hit rates (2/11 = 18%), highlighting the need for ensemble docking approaches that sample multiple conformations.

5. Open Challenges

Several critical gaps limit the current utility of AlphaFold2 for drug discovery. First, induced-fit modeling remains computationally expensive and not routinely applicable at proteome scale. Second, prediction of protein-ligand co-complexes (holo structures) is an unsolved problem; recent methods like RoseTTAFold All-Atom show promise but have not been benchmarked at scale.

Third, post-translational modifications (phosphorylation, ubiquitination) that alter binding site geometry are not captured. Fourth, protein dynamics – especially loop flexibility in active sites – requires molecular dynamics simulation that does not scale to proteome-wide analysis. Integrating MD-generated conformational ensembles with deep learning binding prediction represents a major open research direction.

6. Conclusion

AlphaFold2 structures are broadly useful for structure-based drug discovery, particularly for high-confidence globular domains. The 89 cryptic sites identified in this work represent novel therapeutic opportunities. Closing the performance gap for flexible, disordered, and allosteric targets will require advances in conformational ensemble generation and induced-fit docking at scale.

References

Jumper J. et al. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature* 596. Friesner R.A. et al. (2004). Glide: A New Approach for Rapid, Accurate Docking. *J. Med. Chem.* Le Guilloux V. et al. (2009). Fpocket: An open source platform for ligand pocket detection. *BMC Bioinformatics*. Baek M. et al. (2021). Accurate prediction of protein structures and interactions using RoseTTAFold. *Science*. Binder J.L. et al. (2022). AlphaFold illuminates half of the dark human proteome. *Nat. Struct. Mol. Biol.*